1 - introduction summary

Fortgeschrittene Themen des Softwaretests (Technische Universität München)



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Purpose of Testing

- **What does software testing aim to compare?**
  → *Actual behavior vs. intended behavior.*
- **What's the goal of software testing?**
  → *To show intended functionality is correctly implemented and detect failures.*
- **Is testing a verification or validation method?**
  → *It is a verification method – it checks if the system is built right.*

Test Cases

- **What is a test case?**
  → *Input + expected output + environmental conditions.*
- **What makes a test case "good"?**
  → *High failure detection with good cost-effectiveness.*

Defects & Errors

- **What are the three key types of defects?**
  - *Failure:* I/O mismatch
  - *Error:* State mismatch
  - *Fault:* Root cause in code
- **What does "Defects follow a logarithmic distribution" imply?**
  → *80% of the defects come from 20% of the modules.*
  → *Probability of more faults increases with existing faults.*
- **Why is testing important despite known defect density?**
  → *Because software complexity is high and defects are common.*
- **What is NOT the focus of the testing?**
  → *Debugging, fault localization, and static analysis.*
- **According to statistics, what phase presents the most faults and needs more effort to fix?**
  → *Requirements phase.*

Test Strategy & Selection

- **What is the main goal in test selection?**
  → *Partitioning input data into equivalence classes.*

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- **What is equivalence class testing?**
  → Grouping inputs that should behave similarly to reduce redundant testing.
- **What are alternative test selection strategies?**
  → Structural (control/data flow), stochastic (random), and combinatorial testing.
- **What is limit testing?**
  → Testing at boundaries like MININT, MAXINT, or empty inputs.
- **Why is random testing often used for comparison?**
  → It is simple and acts as a baseline to compare more structured methods.
- **Why is it hard to know if a test strategy is good?**
  → Real systems vary, and outcomes depend on many factors.
- **Why partition-based testing is not always superior to other strategies?**
  → We do not know beforehand which blocks are problematic.
- **What is the main challenge in stopping testing?**
  → Defining meaningful stopping criteria that are not used as performance targets (Goodhart's Law).

Practical Considerations

- **Why shouldn't programmers test their own code?**
  → Psychological bias and organizational issues.
- **Why is test automation critical in practice?**
  → It's essential for continuous integration and regression testing.
- **Why is test selection not everything?**
  → There are also test automation and test management aspects.
- **Name some domains where specifications for testing (expected output) are hard to find:**
  - *Neural networks that distinguish cats from dogs*
  - *NLP for identifying hate speech or misogyny*

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